

Requirements-to-Tests Traceability

RubricGuidedEssayExam — SRS v1.1 mapping to pytest.

Repository: <https://github.com/ALGeek01/RubricGuidedEssayExam>

Tests: MOCK_LLM=1; isolated SQLite (tests/conftest.py).

Test inventory

conftest.py — DB + MOCK_LLM + TestClient + reset

general/test_unit.py — unit

general/test_api.py — integration

security/test_http.py — security

SRS requirement → tests (with explanations)

Requirement	Test(s)	Coverage note	Explanation
§2.0 Essay gen & eval	test_full_exam_single_question_flow; test_two_question_flow	Mock LLM E2E; not live Together	Core product intent: the system must generate essay-style questions at runtime (not only from a fixed bank) and evaluate students' essay answers using the LLM-assisted pipeline. Together these behaviors prove the “oral-style exam in software” goal.
§2.0.1 / FR-GEN-1	—	Not covered	Architecture requirement: English prompt templates in Python are filled at runtime and sent to the LLM; the model returns Python source that assigns a structured object to a named variable; the host executes that code only under sandbox or restricted-exec rules, then validates against schema exam_gen_v1. Tests do not exercise this path when MOCK_LLM is on.
FR-PROF-1	test_exam_start_*; test_full_exam_*; test_two_question_flow	Partial: form fields only	Professors must be able to define and persist an exam configuration: scope/title, number of questions, time limits, domain briefing text, optional reference material, rubric style, and per-question weights summing to 1. The app currently exposes a subset via the start form (e.g., student

			id, domain text, count, education level).
FR-PROF-2 — FR-PROF-4	—	Not covered	FR-PROF-2: at least three independent constraint fields per question slot passed into generation. FR-PROF-3: accommodation multipliers (1.0×–3.0×) on per-question and whole-exam time limits. FR-PROF-4: vary_questions flag so prompts discourage repeating recent questions for the same config.
FR-GEN-2 — FR-GEN-4	—	Not covered	FR-GEN-2: each generated item must include background sheet, essay question, rubric (≥ 3 criteria with points), expected concepts, and metadata. FR-GEN-3: on validation or API failure, retry up to three times with backoff. FR-GEN-4: p95 generation time ≤ 45 s under nominal staging conditions.
FR-STU-1	test_exam_start_*; test_completed_session_*; test_answer_invalid_session	Session + duplicate answer 400	Each exam session is bound to one student and one exam configuration. After the session is finished, closed, expired, or finally submitted, the system must not accept further answers. Tests cover creation, redirect after completion, and rejecting a second answer POST with HTTP 400.
FR-STU-2	—	Not explicitly tested	Questions are shown in professor-defined order or in a randomized order chosen at session start and kept stable across refresh and reconnect for that session.
FR-STU-3	test_full_exam_*; test_two_question_flow	Short answers only	Students submit essay answers as UTF-8 text with a maximum length (SRS: 20,000 characters) enforced on client and server. Current tests use short strings only; boundary and encoding tests are still needed.
FR-STU-4 — FR-STU-5	—	Not covered	FR-STU-4: autosave drafts on a timed and debounced schedule so work is not lost. FR-STU-5: optional per-question and whole-exam timers with server-authoritative expiry (± 5 s) and visible countdown where applicable.
FR-GRADE-1 / FR-GRADE-2	test_full_exam_single_question_flow	Should improve; no schema validation	FR-GRADE-1: the grading prompt must include question, rubric, background, student answer, optional time-on-question, and weighting preset. FR-GRADE-2: the LLM returns a graded structure (exam_grade_v1) via the same

			execute-and-validate pattern as generation, including scores, satisfactory flag, explanation, and confidence.
FR-GRADE-3	—	Not covered	If grading fails after retries, the item is marked needs_human_review and no numeric final is stored until a professor resolves it.
FR-FU-*	—	Not covered	When an answer is not satisfactory and follow-ups are enabled, the LLM may issue follow-up questions and rubrics (up to a max count), then produce a final grade using the initial answer and all follow-up responses. State P is the persisted terminal bundle for that question.
FR-DB-1 / FR-DB-2	(implicit)	No constraint/retention tests	FR-DB-1: store each finalized question under a unique key (student, session, question index) with indexes for session lookup. FR-DB-2: configurable retention window and automated purge of expired records.
FR-FINAL-*	test_two_question_flow	Partial only	When all questions are resolved, aggregate summaries per question and call the LLM for a final percentage and narrative (exam_final_v1). Support at least two professor-selectable schemes (e.g., pure weighted average vs. LLM-adjusted composite). Tests only assert a multi-question path reaches results.
FR-DISP-*	test_full_exam_* (results)	No disputes	FR-DISP-1: after finalization, show the student final percentage, per-question scores, and short explanations. FR-DISP-2/3: optional dispute submission after a cooling-off window and LLM triage with professor-controlled auto-regrade.
FR-REVIEW-1 / FR-REVIEW-2	test_professor_dashboard_and_detail	Smoke only	FR-REVIEW-1: professors filter and list sessions with acceptable query performance on a reference dataset. FR-REVIEW-2: one consolidated view of prompts (redacted secrets), rubrics, backgrounds, student answers, model outputs, timings, and aggregates.
FR-REVIEW-3	—	Not covered	Export a session for archival or accreditation, at minimum as schema-valid JSON (PDF optional).

FR-SEC-1 / FR-SEC-2	—	Not covered	FR-SEC-1: LLM-generated Python must run only in a sandbox or restricted namespace with resource limits. FR-SEC-2: authenticate professor routes; bind student routes to session tokens tied to student and exam config with bounded lifetime.
G-1	—	Not covered	Measurable variation goal: when vary_questions is enabled, at least 90% of questions are not byte-identical to another student's in the same configuration within the prior seven days (stochastic acceptance criteria in SRS).
NFR-PERF-1; NFR-REL-1; NFR-AUDIT-1	—	Not covered	NFR-PERF-1: student UI performance (e.g., LCP ≤ 2.5 s under a reference network profile). NFR-REL-1: core API availability (e.g., $\geq 99.5\%$ over pilot). NFR-AUDIT-1: append-only audit events with correlation IDs and high durability on commit.
NFR-LOG-1 (partial)	test_*traceback*; test_exam_start_rejects_empty*	No traceback in HTML	Operational logging must omit API secrets; logging full student answers is off by default. Security tests check that common error pages do not embed Python tracebacks or host file paths in HTML.
NFR-ACC-1; NFR-I18N-1; NFR-DOC-1	—	Not covered	NFR-ACC-1: WCAG 2.1 AA on primary student flows with automated axe checks. NFR-I18N-1: externalize UI strings (en-US in v1). NFR-DOC-1: operator runbook for deploy, env vars, backup, and incidents within a page budget.
ADV-CURVE-1; ADV-DISC-1	—	Not covered	Optional advanced features: cohort-based curved grading with minimum sample size and visible fallbacks; or discussion-based testing with bounded transcript length and token budget passed into each grading call.

Unit tests vs SRS (with explanations)

Test	SRS relationship	Explanation
test_education_levels_ids_are_unique	Optional SRS: education/audience level	Catalog tests: each education level has a unique id so the UI and prompts cannot reference ambiguous tiers.

test_label_for_level_*	Optional SRS	Human-readable labels (e.g., “High school (9–12)”) display correctly for known ids and degrade gracefully for unknown ids.
test_guidance_for_level_*	Optional SRS	Prompt guidance text matches the selected difficulty/audience; unknown levels fall back to the default level’s guidance so generation still behaves predictably.
test_together_api_error_message_and_http_status	Supports FR-GEN-3 if wired to retries	TogetherApiError carries a message and HTTP status so higher layers can map provider failures to retries or user-visible errors without losing context.

Summary

Covered: basic exam flow, two questions, 404s, professor smoke, no traceback on errors, duplicate answer rejected.

Gaps: FR-GEN-1 sandbox, FR-SEC-*, timers, schema grading, follow-ups, DB retention, final schemes, disputes, export, most NFRs.